

RICHMOND, CA, USA

WMO: 994036

Lat: 37.923N

Lon: 122.410W

Elev: 6

StdP: 101.26

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	5.7	6.6	N/A	N/A	N/A	N/A	N/A	N/A	10.1	13.5	9.0	12.3	2.9	350	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	5.8	25.0	N/A	23.2	N/A	21.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.2	6.1	4.0	30.2	1.2	2.7	3.1	32.1	2.4	33.6	1.7	35.1	0.8	37.1		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	11.2	11.8	12.9	13.8	14.7	16.0	16.4	16.9	17.8	16.7	13.9	10.9
	DBStd	3.07	2.13	2.16	2.09	2.24	1.96	2.01	1.67	1.84	2.11	2.36	2.08	2.22
	HDD10.0	35	13	6	1	0	0	0	0	0	0	0	1	15
	HDD18.3	1478	220	182	169	137	114	76	65	54	34	61	135	231
	CDD10.0	1653	51	57	90	115	147	180	198	213	234	209	117	43
	CDD18.3	53	0	0	0	2	3	6	4	8	18	12	1	0
	CDH23.3	149	0	0	1	12	13	27	7	13	40	35	2	0
	CDH26.7	25	0	0	0	1	4	7	0	1	7	5	0	0
Wind	WSAvg	2.9	3.0	3.0	2.8	3.2	3.2	3.2	3.2	3.0	2.5	2.4	2.6	2.8
Precipitation	PrecAvg	626	129	125	69	40	24	4	0	2	1	30	61	138
	PrecMax	1022	339	407	234	149	91	22	1	24	7	116	196	341
	PrecMin	302	8	0	0	5	0	0	0	0	0	0	0	11
	PrecStd	183	94	87	69	35	24	6	0	5	2	30	56	97
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	20.9	21.8	25.2	24.7	27.5	24.2	25.2	27.3	26.7	22.8	18.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	17.2	17.8	18.8	21.0	21.1	22.7	21.7	22.7	24.0	24.0	20.1	16.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	15.5	16.1	17.2	18.7	19.0	20.6	20.3	21.1	22.3	21.6	18.0	15.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	14.1	14.9	15.8	16.9	17.7	19.2	19.1	19.9	20.8	19.9	16.7	14.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.0	5.4	5.9	5.6	5.9	5.3	5.4	5.8	5.6	5.0	4.3
		MCDBR	7.1	7.1	7.6	8.5	8.2	8.7	7.0	7.2	8.1	8.6	7.2	5.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.322	0.327	0.338	0.357	0.362	0.357	0.365	0.345	0.344	0.333	0.321	0.323	
	taud	2.560	2.546	2.505	2.420	2.435	2.454	2.433	2.490	2.481	2.555	2.566	2.561	
	Ebn at Noon	865	907	926	922	918	919	908	923	906	888	859	838	
	Edh at Noon	75	86	98	113	114	112	113	104	100	84	74	71	
All-Sky Solar Radiation	RadAvg	2.36	3.29	4.66	6.12	7.08	7.75	7.37	6.55	5.54	4.05	2.79	2.06	
	RadStd	0.34	0.41	0.43	0.49	0.46	0.34	0.25	0.28	0.21	0.27	0.19	0.31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (46 neighbors)	N/A	N/A	-1.19	N/A	N/A	N/A	N/A	-110	+135	N/A	

Nomenclature: See separate page